Exercise Sheet 3

Strong and Weak Ties

5942UE Network Science

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3.A Triadic Closure

Exercise 3.A.1: Open Triads and Predicted Closures

Consider the following friendship network (all edges are **strong** ties unless stated otherwise):

Edges: A–B, A–C, A–D, B–C, D–E, E–F, D–F.

1. List all **open triads** in the network.
2. For each open triad, which missing edge is predicted to appear by **triadic closure**? Rank them by "closure pressure".
3. Which open triad, if closed, would increase the clustering coefficient of node A the most?
4. Node G joins and forms strong ties with A and D. Predict which edge forms next.

Exercise 3.A.2: Computing Clustering Coefficients

For the graph: A–B, A–C, A–D, B–C, D–E, E–F, D–F:

1. Compute the **local clustering coefficient** C_v for every node.
2. Compute the **average clustering coefficient** $\bar{C}$.
3. Interpret the result: what does it tell us about the network?
4. Visualise the graph with nodes coloured by C_v.

3.B Bridges and Brokers

Exercise 3.B.1: Finding Bridges and Structural Holes

Three cliques: $1, 2, 3$, $4, 5, 6$, and $7, 8, 9$. Bridges: $3–4$ and $6–7$.

1. Is $3–4$ a bridge? Is $6–7$ a bridge?
2. Is $3–4$ a **local bridge**?
3. Node 5 in Clique 2 connects to Cliques 1 and 3. Does it sit in a **structural hole**?
4. If node 3 forms a direct edge with node 7, what happens to the hole and the bridges?

Exercise 3.B.2: Identifying Bridges with NetworkX

Build the three-clique graph in NetworkX. Identify bridges and rank edges by edge betweenness.

3.C Weak Ties Theorem

Exercise 3.C.1: Verifying the Weak-Ties Theorem

Weak-Ties Theorem: Under Strong Triadic Closure (STC), every local bridge is a weak tie.

Check STC and local bridges for:

- **A:** Strong $1–2, 1–3$. Weak $2–4$. No $2–3$.
- **B:** Strong $1–2, 1–3, 2–3$. Weak $1–4$.
- **C:** Strong $1–2, 2–3, 1–3, 3–4, 2–4$.

Exercise 3.C.2: Granovetter's Job Study

1. Why are rarely-seen acquaintances more useful for job finding than close friends?
2. Apply the weak-ties theorem to redundant vs. novel information.
3. Is "unfriending all close friends" good advice based on the theorem?
4. How does LinkedIn fit this framework?

3.D Evidence and Overlap

Exercise 3.D.1: Neighbourhood Overlap Calculation

$$O(u, v) = \frac{|\mathcal{N}(u) \cap \mathcal{N}(v)|}{|\mathcal{N}(u) \cup \mathcal{N}(v)|}$$

(excluding u and v from $\mathcal{N}$)

For edges: A-B, A-C, A-D, B-C, B-E, C-F:

1. Compute $O(A, B)$, $O(A, C)$, and $O(B, E)$.
2. Rank them. Which is most like a local bridge?

Exercise 3.D.2: Neighbourhood Overlap in NetworkX

Compute average neighborhood overlap within and across factions in the karate club.

Exercise 3.D.3: Practical Lab — Weak Ties in the Karate Club

Use `nx.karate_club_graph()` to connect the lecture concepts to measurable graph quantities.

1. Compute the **local clustering coefficient** C_v for every node.
2. Compute the **neighbourhood overlap** $O(u, v)$ for every edge.
3. Classify edges as:
   - **weak / bridge-like** if $O(u, v) < 0.10$
   - **medium** if $0.10 \le O(u, v) < 0.30$
   - **strong / embedded** if $O(u, v) \ge 0.30$
4. Visualise the graph:
   - node colour = clustering coefficient
   - edge colour and width = overlap-based tie strength
5. List the five lowest-overlap edges. Are they mostly within one faction or between factions?

3.E Tie Strength in Practice

Exercise 3.E.1: Tie Strengths in Practice

1. Facebook (symmetric) vs. Twitter (asymmetric): structural implications for overlap?
2. Why do maintained relationships grow slowly while total count grows quickly?
3. How should you design a recommender for **information diversity**?

Exercise 3.E.2: Simulating Tie Strength and Overlap

Assign synthetic strength values to karate-club edges (within-faction strong, across-faction weak) and compute the correlation with neighborhood overlap.